

Difference Between SUM() and COUNT() in SQL

In SQL, both **SUM()** and **COUNT()** are aggregate functions, but they serve different purposes. Let's understand their differences and when to use each.

Feature	SUM()	COUNT()
Purpose	Adds up all numeric values in a column.	Counts the total number of rows or non-NULL values.
Data Type	Works only with numeric data types.	Can work with any data type.
NULL Handling	Ignores NULL values while summing.	Includes NULL values when counting specific columns.
Return Type	Returns the total sum (numeric result).	Returns an integer (count of items).

When to Use Each Function:

- 1. Use SUM():** When you want to calculate the total of numeric data, like total sales, total marks, or total salary.
- 2. Use COUNT():** When you want to find the number of rows or the number of entries in a specific column.

Examples:

Example 1 - Using SUM():

Suppose you have a table named **sales** with a column **amount**.

To find the total sales:

```
SELECT SUM(amount) AS Total_Sales FROM sales;
```

This will return the total of all values in the **amount** column.

Example 2 - Using COUNT():

Suppose you want to count how many sales transactions are recorded in the same table:

```
SELECT COUNT(*) AS Total_Transactions FROM sales;
```

This will return the total number of rows in the **sales** table.

Example 3 - Using COUNT(column_name):

To count only the non-NULL values in a specific column:

```
SELECT COUNT(amount) AS Count_of_Amounts FROM sales;
```

This counts only the rows where the **amount** value is not NULL.

Summary:

- **SUM()** → Adds numeric values.
- **COUNT()** → Counts rows or non-NULL values.
- Use SUM() for totals and COUNT() for the number of items.